Lightest full-fidelity feasible design per gas, u_res_max = 0.5 m/s — impact transient

Horizontal Displacement

x [m]

Horizontal Velocity

u_x [m/s]

Horizontal Acceleration

a_x [m/s^2]

-A_LIM

Horizontal Force

F_x [N]

Vertical Penetration

y_pen [m]

Displacement Along n

Vertical Velocity

u_y [m/s]

Velocity Along n

Vertical Acceleration

a_y [m/s^2]

-A_LIM

Bag Load Along n (F_n/M)

Net Vertical Force (incl. g)

F_y,net [N]

Pressure Force Along n

h [m]

dotted: D0 of each design

u_n [m/s]

F_n/M [m/s^2]

A_LIM

F_n [N]

Gas Mass

m [kg]

Time [s]

Airbag Volume

V [m^3]

Time [s]

Internal Pressure

1e5

P_bag [Pa]

Time [s]

Gas Temperature

T [K]

Time [s]

N2: cylinder / Nylon, m = 0.583 kg

He: cylinder / Nylon, m = 0.379 kg

Ar: cylinder / Nylon, m = 0.732 kg

CO2: cylinder / Nylon, m = 0.707 kg

dry air: cylinder / Nylon, m = 0.591 kg